Business 41204

Machine Learning

Causal Inference

Outline:

1. Introduction
2. Basic Causal Setup and A/B Tests
3. Heterogeneous Effects and Targeting
4. Observational Data and (De-) Confounding
5. Summary

1. Introduction

Data and ML

Data are only informative about the patterns that exist in the data

Supervised/unsupervised learning about finding these patterns

Patterns in data

- provide information about population represented by data
- useful for answering descriptive questions:
  - E.g., what is average historical spending among customers of different (observed) types from the population represented by the data?
- useful for answering prediction questions:
  - E.g., if I see an observation with characteristics x from the population represented by the data, what do I guess the associated y will be?

Causal Questions

Many business questions are fundamentally *causal*:

- If I *send* a promotion to a customer, will that make it more likely the customer buys my product
- If I *set* my price to *x*, how will that impact my sales?
- If I *change* my website to display fewer product recommendations, will that increase click through on the recommended products?
- If someone *takes* a drug, will it decrease symptoms of a disease?
- …

All of these questions involve *intervening* in a system

- Asking what happens if we change some specific feature of the system (without altering other features/things outside of our control)
- Without *outside/auxiliary* input, there is no information about causal questions in data
  - “correlation does not imply causality”

Simple Example

I could use ML to figure (generalized) versions of this out.

Suppose you have data on two variables:

- A = 1 if a person took an aspirin in a given period (and 0 otherwise)
- P = 1 if a person reported being in pain over the same period (and 0 otherwise)

A and P are highly (positively) correlated in data. What does this mean?

- If I see $A = 1$ but don't see P, I should predict $P = 1$
- If I see $P = 1$ but don't see A, I should predict $A = 1$
  - These predictions will be pretty good!

What does this tell me about *causality* between A and P?

- NOTHING!
  - A could cause P
  - P could cause A
  - Some other factor could cause A and P

A more subtle question: What if I see two variables that are not associated in data? Does this mean there is no causal relationship between them?

Code for churn causality example

A More Realistic Example

Let's look at a subscriber retention (churn) example:

- Outcome: Renew Subscription (0/1)
- Predictors
  - Discount – fractional discount offered to customer upon renewing
  - Ad spend – ad money spent targeted at this user
  - Monthly usage – percent of days last period user was active
  - Last upgrade – time since last product upgrade
  - Bugs reported – number of bugs reported by user
  - Interactions – number of interactions with customer
  - Sales calls – number of sales calls made to customer
  - Economy – state of the customer's regional economy, normalized to 0 (worst)-1 (best)

- Train a predictive model using boosted trees
  - Tuning parameters in code

Adapted from this article

Aside: SHAP Values

We will use SHAP values as our interpretation device in this example

SHAP (SHapley Additive Explanations)

- Popular interpretation device motivated by cooperative game theory (Shapley, 1953)
- Basic idea is want to decompose group payoff/output into marginal contribution of each player
- Can think about input variables as players and prediction as payoff

Aside: SHAP Values

Let $v(S) = E[f(X)|X_S]$

- $f(X)$ is prediction model
- X_S is a subset of the predictor variables
- $v(S)$ is thus expected output given a subset of predictors

SHAP value for feature j:

$$\phi_j = \sum_{S \subseteq \{1,\ldots,p\}/j} \frac{|S|!\,(p - |S| - 1)!}{p!} (v(S \cup j) - v(S))$$

- Sum is over all combinations of predictors that do not include feature j
- SHAP value is effectively average marginal contribution of j across all possible models
- Hard to compute outside of simple examples. Shortcuts are typically made. (Not important for us)

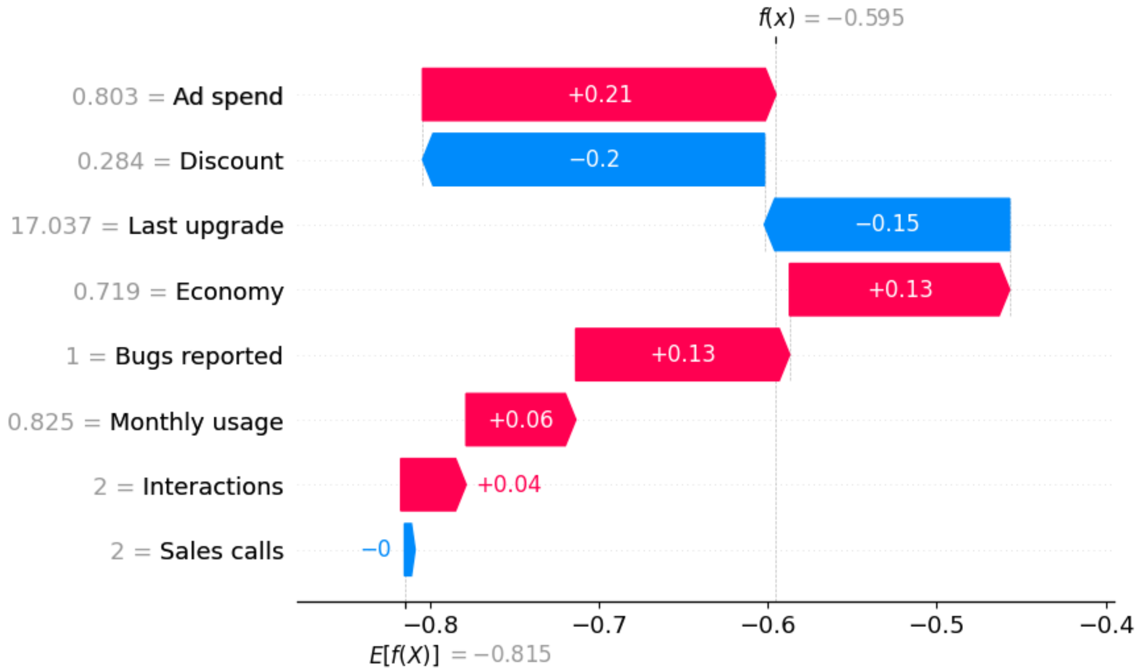
Aside: SHAP Values

For a given prediction $f(x_i)$, SHAP have the property

$$f(x_i) = E[f(X)] + \sum_{j=1}^{p} \phi_j(x_i)$$

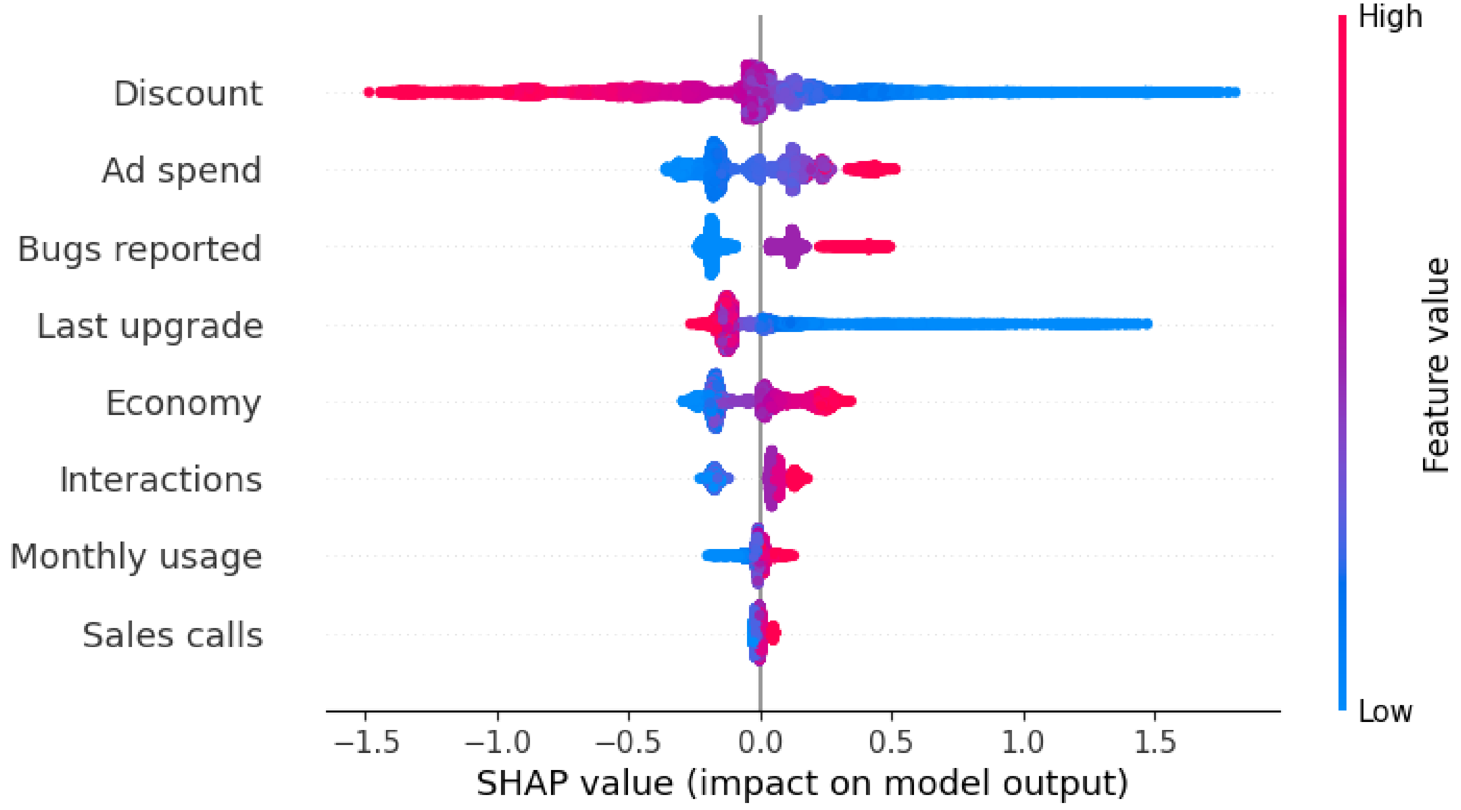
- The predicted value is the overall average plus the sum of SHAP across all predictors
- SHAP differ for every different possible value of X

SHAP for first observation:

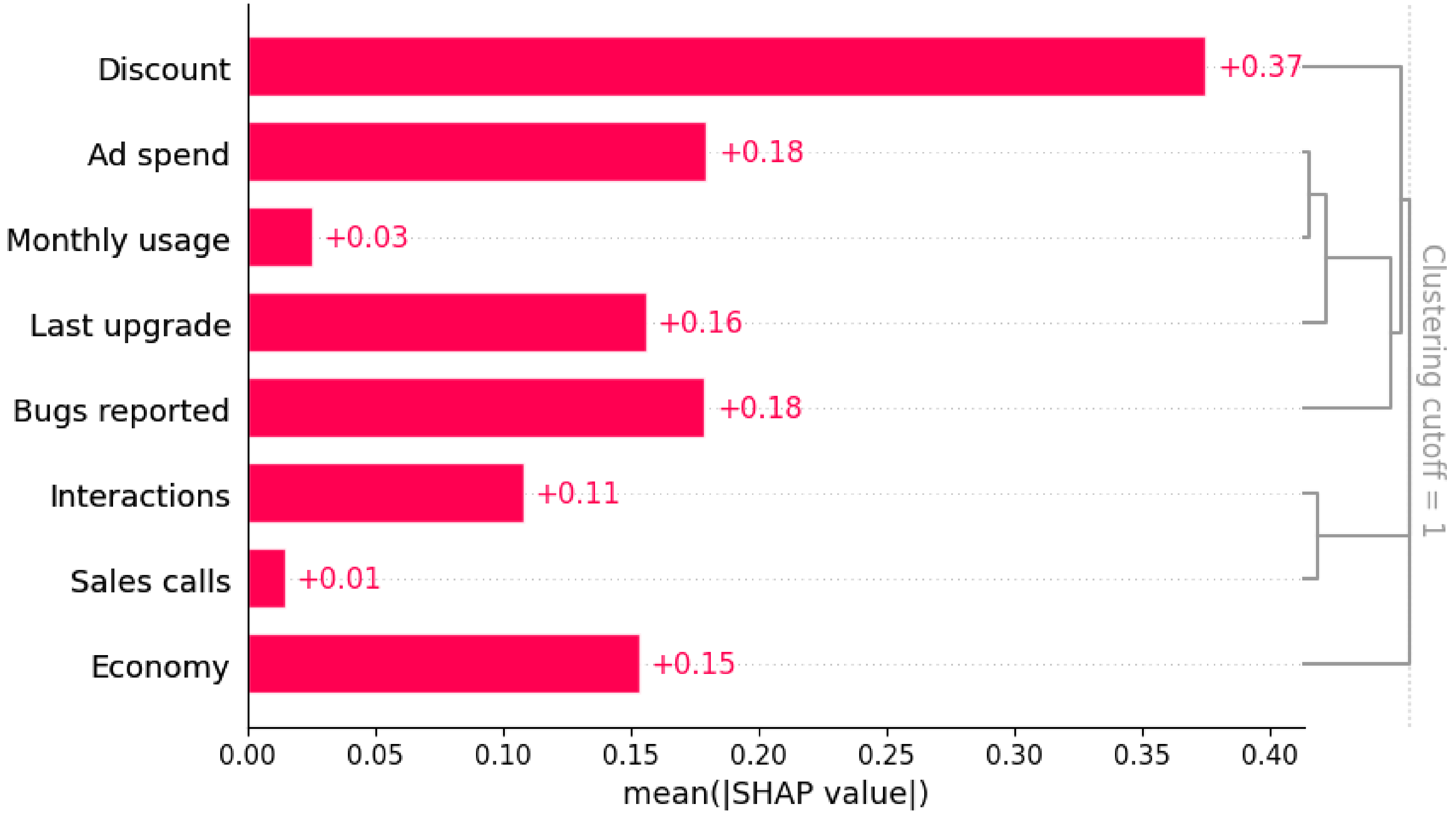


SHAP Values

SHAP for observations in our example:



SHAP Variable Importance



Several variables “look” important.

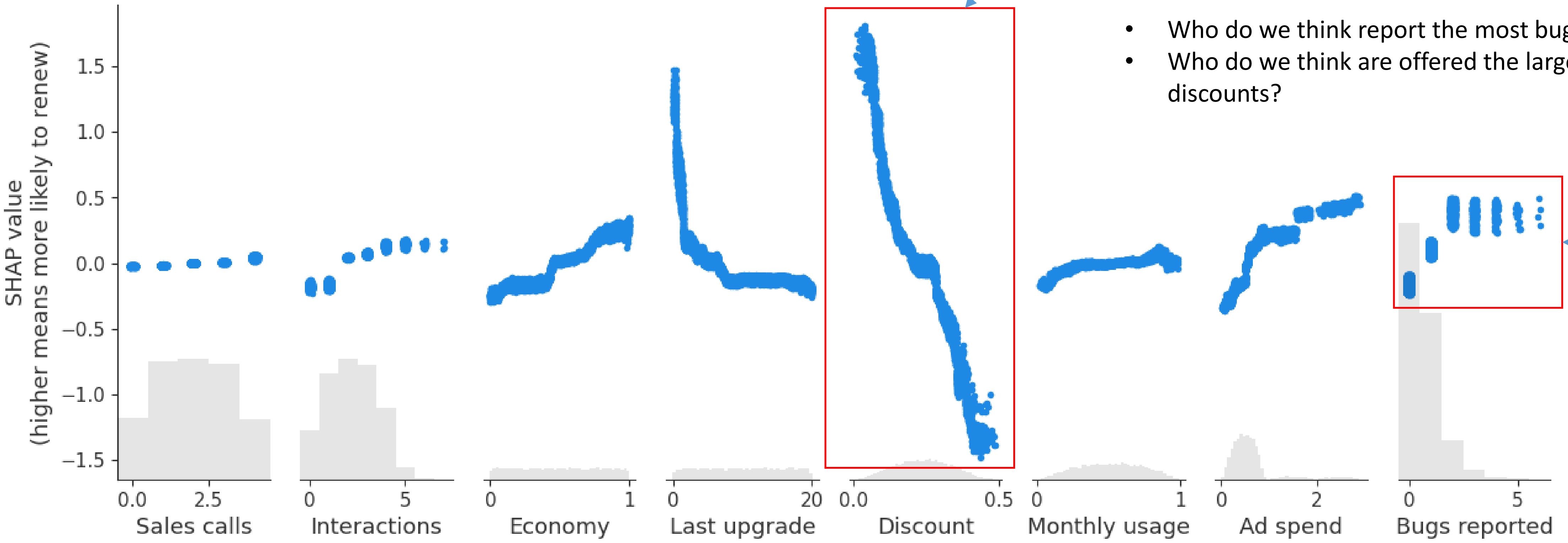
Nothing seems crazy here.

The “clustering” is trying to capture “redundant” features.

- Redundant roughly means prediction quality doesn’t change w/ one vs other
- Higher values mean less redundant
  - 0 means no difference in prediction quality

Variable Interpretation

SHAP values across features (larger values mean more likely to renew):



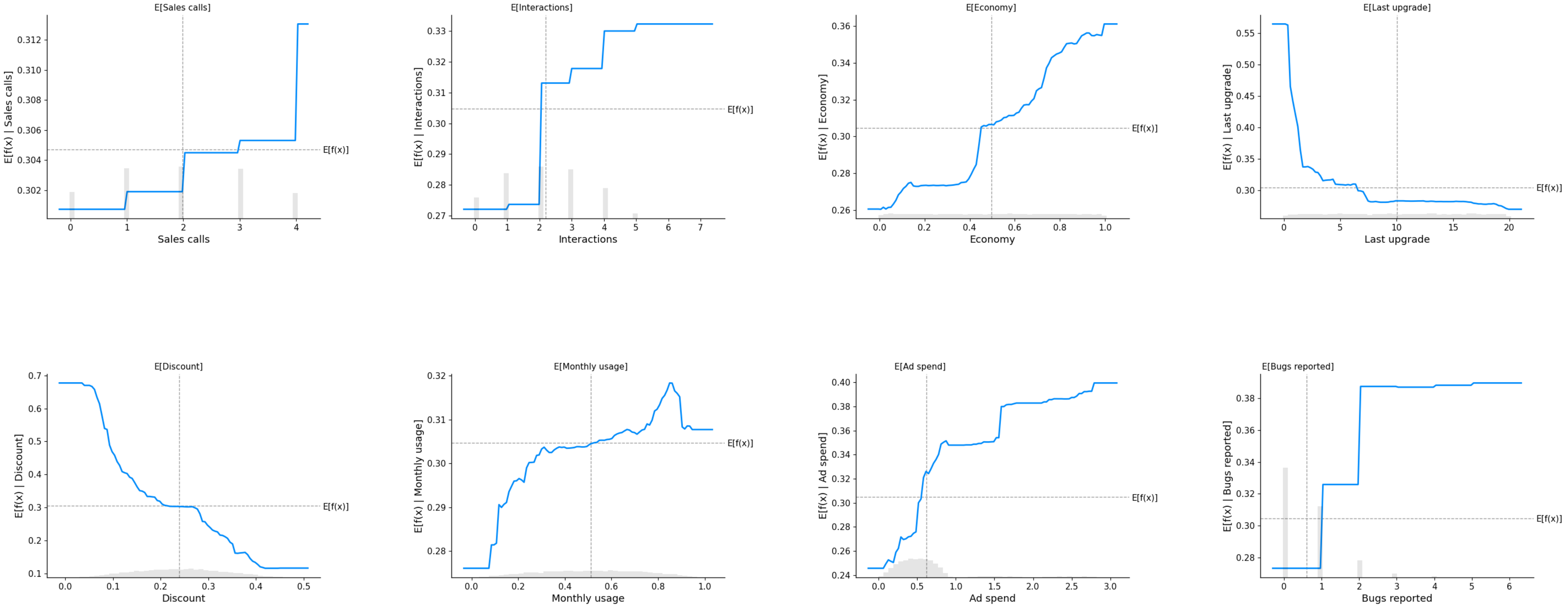
Predict people with low discounts and reporting lots of bugs more likely to renew

- Should we just discontinue discounts?
- Should we make our product more buggy?

- Who do we think report the most bugs?
- Who do we think are offered the largest discounts?

What are we learning about here?

Partial Dependence Plots



ML and Causality

Supervised/unsupervised learning about finding patterns in data

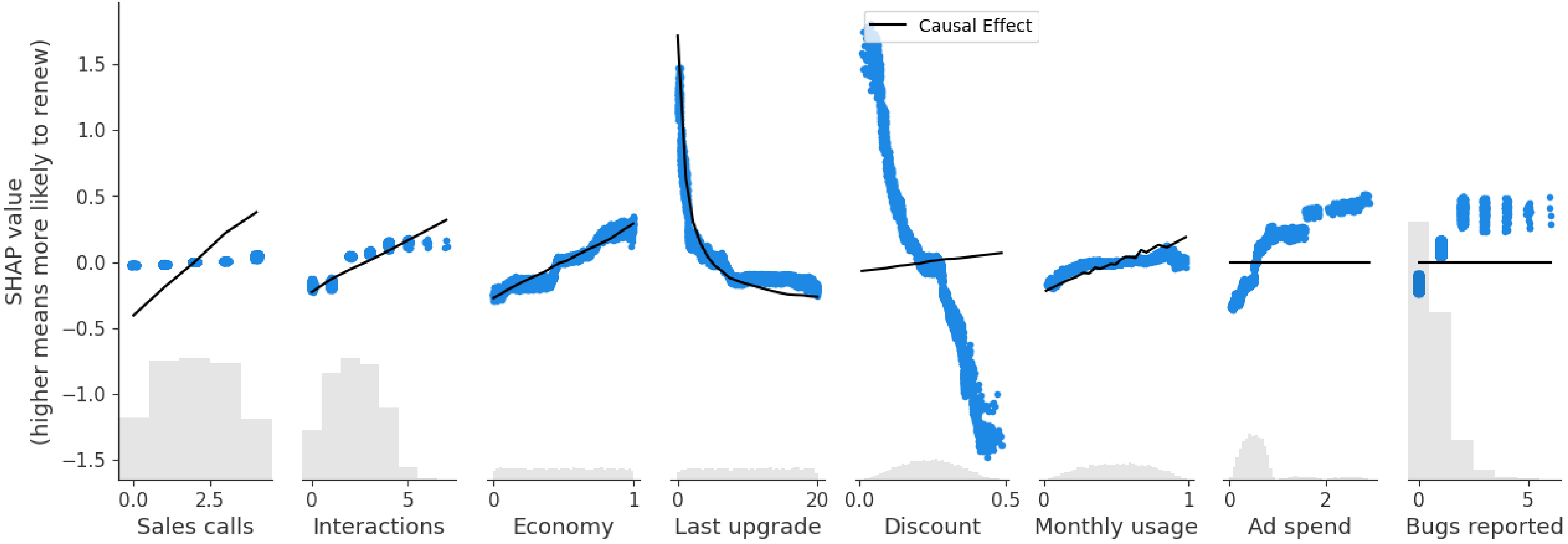
- Patterns in data useful for lots of things
- By themselves, give NO information about causality
- Be wary of anyone who recommends interventions on the basis of “predictive analytics” or any other ML/purely data-driven method

Let’s think a little more about our renewal example:

- What is our goal?
  - Provide a forecast of customer retention for revenue/financial planning assuming business as usual?
    - Perfect! That’s just what our model is designed to do.
  - Decide what action/intervention to make to change away from business as usual to retain more customers?
    - Need to be very careful here. Observational data only tells us about the environment under observation
    - Intervening (if the intervention matters) changes this environment
    - Need a mechanism to understand the changes

Variable Interpretation

- SHAP values across features
- True causal effect shown by black line
  - Know these because the data are fake and we know how they were generated
  - In real life, you don't know the true effects



For some variables the causal effects are very different from the predictive effects!!

ML and Causality

How do we find causal relationships then?

- We use auxiliary information from outside of the data!
  - Ideal case: Know data were generated as part of an experiment
    - Being able to (randomly) assign treatment/intervention means we know flow of causality
    - Intuitively, gives us information under “usual” and “intervened” environment
  - Otherwise: Make assumptions about causal channels
    - E.g. use “models”
    - Only as good as the assumptions, but often seems more reasonable than just taking raw patterns in data seriously

How does ML fit in?

1. With experimental data, ML can help us understand heterogeneity and improve policy targeting
2. With observational data and a “model”, ML can help us account for confounding factors

- In both cases, ML is supplementary

2. Basic Causal Setup and A/B Tests

Fundamental Problem of Causal Inference

Consider a case with a binary *treatment* ($T \in \{0,1\}$)

- Take a pill (1) or not (0)
- Receive a promotion (1) or not (0)
- See advertisement A (0) or advertisement B (1)

Define outcomes

- $Y(1)$ outcome of unit when $T = 1$
- $Y(0)$ outcome of unit when $T = 1$

Ideally, would like to know *treatment effect:* $Y(1) - Y(0)$

- Corresponds to the exact effect of the treatment for a specific unit

BUT, impossible to observe both $Y(1)$ and $Y(0)$

- For units with $T = 1$, see $Y(1)$
- For units with $T = 0$, see $Y(0)$
- Implies impossible to learn $Y(1) - Y(0)$

Summaries of Causal Effects

Rather than try to learn $Y(1) - Y(0)$ (which is impossible), try to learn summaries.

Average Treatment Effect:

$$ATE = E[Y(1) - Y(0)]$$

- Expected effect of the treatment if given to random unit in the population

Conditional Average Treatment Effect:

$$CATE(x) = E[Y(1) - Y(0)|X = x]$$

- Expected effect of the treatment if given to random unit that has characteristics x
- Useful for targeting treatment – e.g. targeted marketing
- Sometimes referred to as individual treatment effect – which is misleading

Problem with Observational Data

What can we actually observe?

- T: whether or not unit has the "treatment"
- Y: observed outcome for the unit
  - $Y = TY(1) + (1 - T)Y(0)$
  - I.e. the observed outcome is $Y(1)$ for units with the treatment and $Y(0)$ for units without the treatment

Suppose we want to learn the *ATE* ($E[Y(1) - Y(0)]$):

- Can learn
  - $E[Y|T = 1] = E[Y(1)|T = 1]$ (average outcome among units with $T = 1$)
  - $E[Y|T = 0] = E[Y(0)|T = 0]$ (average outcome among units with $T = 0$)

- If we knew $E[Y(1)|T = 1] = E[Y(1)]$ and $E[Y(0)|T = 0] = E[Y(0)]$, we'd be done!

Problem with Observational Data

If we knew $E[Y(1)|T=1] = E[Y(1)]$ and $E[Y(0)|T=0] = E[Y(0)]$, we'd be done!

But what do these formulas actually mean?

- $E[Y(1)]$ is the average outcome (under treatment) of a random unit from the overall population
- $E[Y(1)]|T=1]$ is the average outcome (under treatment) of a random unit from the population of units that "***chose***" to have the treatment
- IN GENERAL, THESE WILL BE DIFFERENT! The difference is termed *selection bias*
  - E.g. people who use my product more are more likely to report bugs (T) and more likely to renew their contract (Y)

Retail Spending Example

Code for spending causality example

Let’s look at an example

Outcome variable:

- Spending: Total spending by customer over a specific three month window

A data science team has built predictive models for spending and found that whether a customer has a consumer credit card is a very strong predictor of spending

- Average spending|has credit card: 19.00
- Average spending|no credit card: 7.02
- Difference: 11.99 (standard error: 0.49)

The data science team then suggests that the company prioritize getting people to sign up for the consumer credit card to increase customer spending (and firm revenue)

Is this a good idea?

Experiments

Experiments – aka A/B tests, randomized controlled trials (RCTs) – allow us to learn causal effects!

Recall that we can easily learn

- $E[Y|T=1] = E[Y(1)|T=1]$ (average outcome among units with $T=1$)
- $E[Y|T=0] = E[Y(0)|T=0]$ (average outcome among units with $T=0$)

What does an experiment buy us?

- Intuition is clear: *If we assign the treatment (randomly), people don't get to "choose" whether they are treated*

- Means $E[Y|T=1] = E[Y(1)|T=1] = E[Y(1)]$ and $E[Y|T=0] = E[Y(0)|T=0] = E[Y(0)]$
    - Effectively, expected outcome among observed treated (control) observation is expected treated (control) outcome of *random* unit because treatment was assigned randomly

Can learn, e.g., *ATE* as

$$ATE = E[Y(1) - Y(0)] = E[Y|T=1] - E[Y|T=0]$$

Retail Spending Example

Let's go back to the retail spending example

Treatment variable:

- Promotion: A binary variable indicating that the customer was randomly assigned to receive a promotion

Outcome variable:

- Spending: Total spending by customer over the three month window *following* random assignment of Promotion

Let's look at average spending among people who were randomly assigned to receive the promotion and those who weren't:

- Average spending|promotion: 10.11
- Average spending|no promotion: 7.15
- Difference: 2.96 (standard error: 0.299)

Should we think about rolling out this promotion more broadly if our goal is to increase 3 month spending?

- Are there other things we should think about?

3. Heterogeneous Effects and Targeting

Heterogeneous Effects

In many cases, interest is not in an overall summary (like the *ATE*) but in understanding differences in treatment impact across units

- *Treatment effect heterogeneity*

The CATE tells us the average effect of treatment among units with shared characteristics

- Recall $CATE(x) = E[Y(1) - Y(0)|X = x]$
- Understanding *CATE* can be interesting in its own right
- Understanding *CATE* important input into targeting
- ML methods useful for learning *CATE* and targeting rules!

Estimating CATE

Note that

$$CATE(X) = E[Y(1) - Y(0)|X] = E[Y|T = 1, X] - E[Y|T = 0|X]$$

Estimating *CATE* reduces to estimating prediction model for Y:

1. Can directly build prediction model for Y using predictors T, X
   - Usually not recommended because small treatment effects can be swamped by highly predictive X's
2. Subset data into $T = 1$ and $T = 0$ subsets. Build predictive model in each subset.
   - Emphasizes that learning differences across T is the goal. Does not directly target *CATE* though
3. Less obvious solution: (i) Construct

   $$\tilde{Y} = \left(\frac{T}{p} - \frac{1-T}{1-p}\right)\left(Y - g(T, X)\right) + g(1, X) - g(0, X)$$

   where p is the treatment probability and $g(T, X)$ is an estimated prediction rule for Y given T and X.
   (ii) Build model to predict $\tilde{Y}$ with X
   - Works because $E\left[\tilde{Y}\middle|X\right] = E[Y(1) - Y(0)|X]$. I.e. directly targets *CATE*

Neither 1, 2, or 3 is uniformly superior.

- Any supervised ML method can be used in the prediction tasks
- In all cases, can choose tuning parameters/models on the basis of validation performance for predicting Y or $\tilde{Y}$
- Getting good predictions for Y or $\tilde{Y}$ not equivalent to getting good estimates for *CATE*

Retail Spending Example

Let's go back to the retail spending example

- Estimate heterogeneous effects of the promotion
- Consider approach based on predicting $\tilde{Y}$
  - Tried lasso, regression trees, and random forests (with some tuning) to form $g(T, X)$
    - Lasso did the best according to CV MSE, so all results are based on using the lasso model
- Consider three learners for the CATE
  - Lasso
    - CV MSE: 12950, Validation MSE: 8891
  - Regression Tree
    - CV MSE: 13150, Validation MSE: 8904
  - Random Forest
    - OOB MSE: 12937, Validation MSE: 8893

Interpreting Estimated CATE

Sometimes interested in understanding features related to different responses to treatment

Any interpretation device from supervised learning can be used to aid in interpreting the estimated CATE

Spending example, just look at lasso model and random forest:

- Lasso: just look at variables selected and their coefficients
- Random Forest: VI and SHAP

Retail Example: Lasso Estimates

For the CATE model:

- Select only five variables

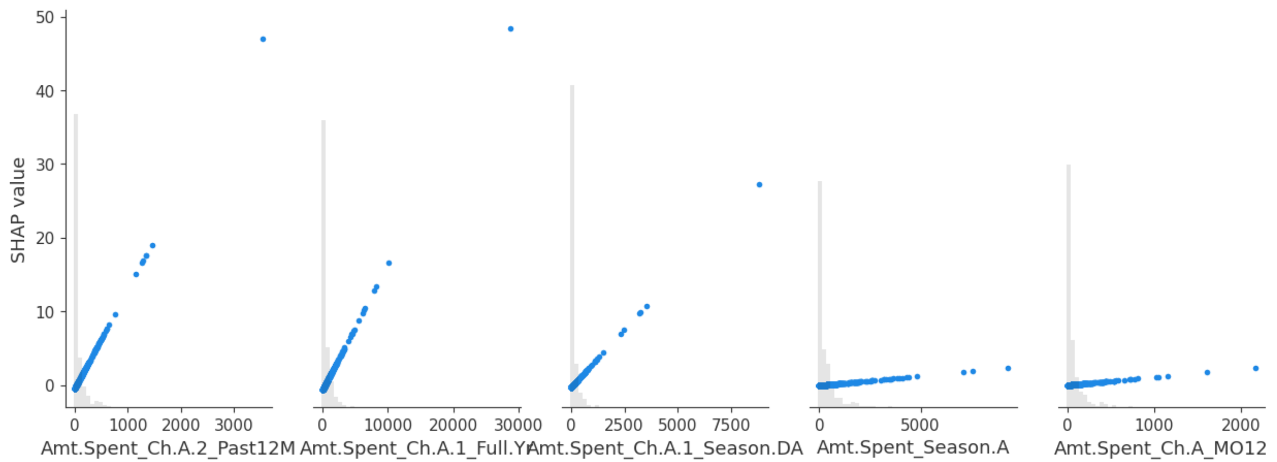
  Amt.Spent – self explanatory

  Ch.? – how the consumer is engaging

  Season.? – Season of year

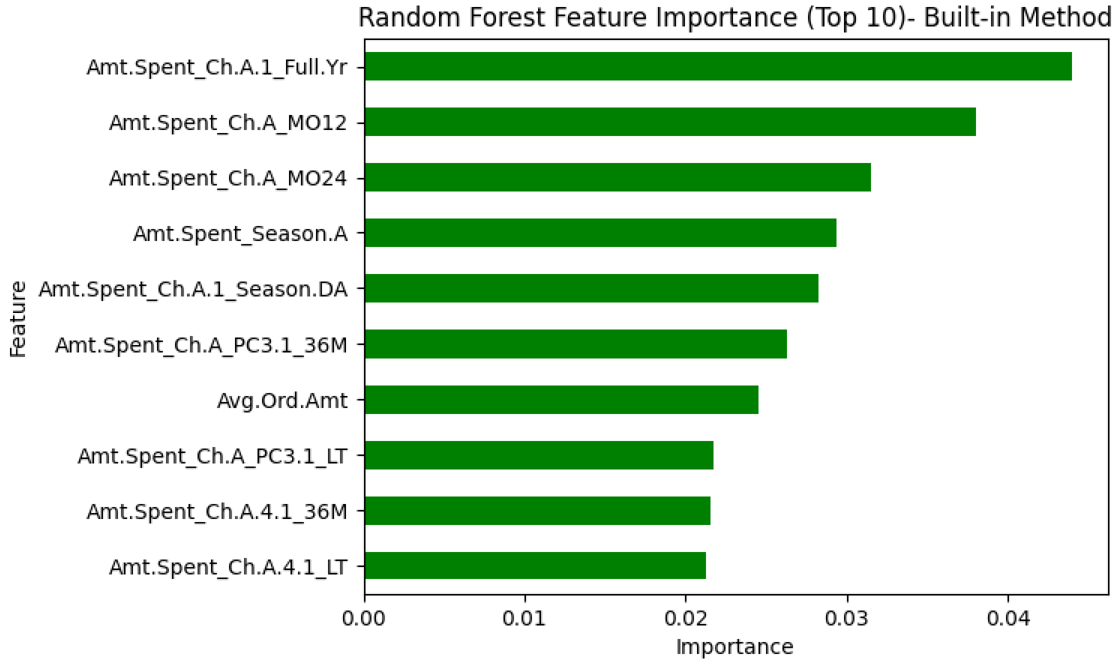
Feature	Coefficient
Amt.Spent_Ch.A.1_Full.Yr	1.545890
Amt.Spent_Ch.A.1_Season.DA	0.800846
Amt.Spent_Ch.A.2_Past12M	2.406343
Amt.Spent_Ch.A_MO12	0.183957
Amt.Spent_Season.A	0.207745

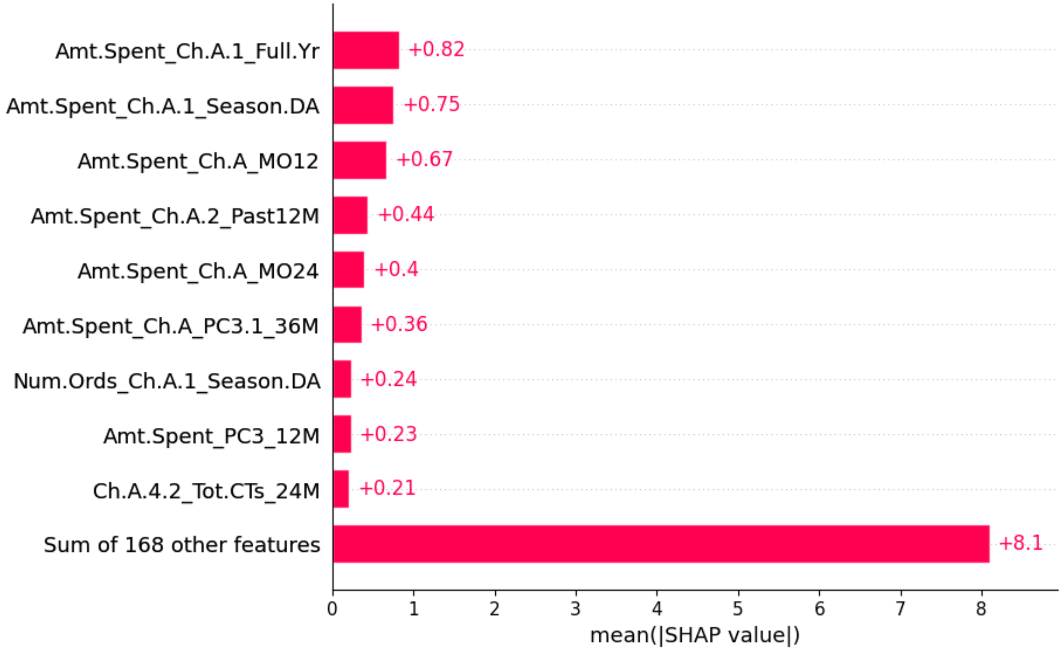
Do we think these are the “right” variables?



Retail Example: Random Forest Estimates

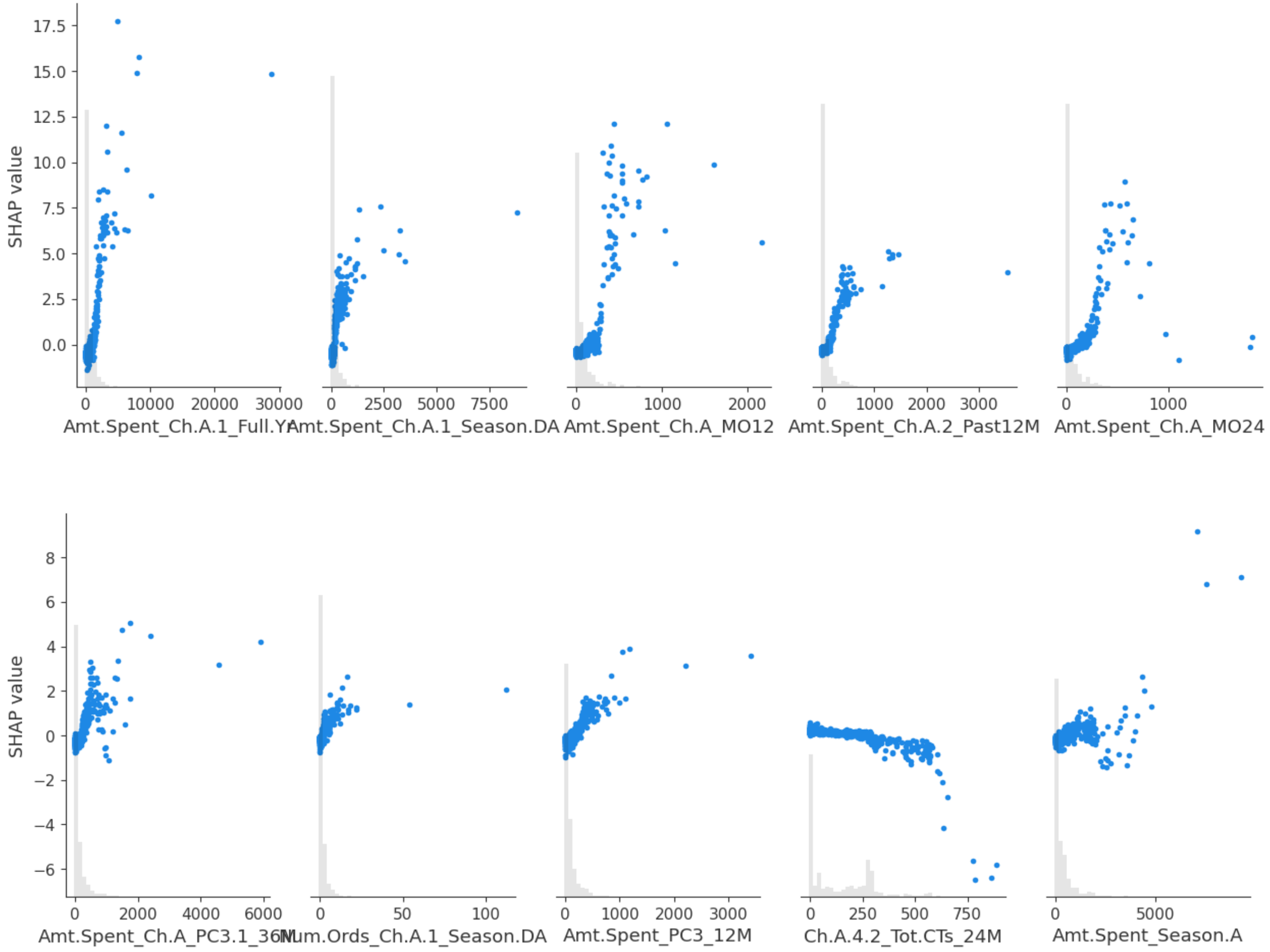
Built-in and SHAP variable importance measures for random forest CATE estimates:





Retail Example: Random Forest Estimates

Visualization of SHAP values for random forest model:



Targeting

One of the chief reasons we care about *CATE* is to learn whom to target:

- If treatment is costless, would like to target individuals with $CATE(x) \geq 0$
- More generally, target individuals with $CATE(x) \geq Cost(x)$ where $Cost(x)$ is the cost of treating units with characteristics x

With estimated *CATE,* can estimate *policy function*

$$\pi(x) = 1(CATE(x) \geq Cost(x))$$

Value of Targeting

Rather than evaluate *CATE* models on the basis of predicting Y or $\tilde{Y}$, can evaluate on "profits" from following implied policy:

$$
\begin{aligned}
V(\pi) &= E\Big[\pi(X)\big(Y(1) - Cost(X)\big) + \big(1 - \pi(X)\big)Y(0)\Big] - E[Y(0)] \\
&= E\Big[\pi(X)\big(Y(1) - Y(0)\big)\Big] - E[\pi(X)Cost(X)] \\
&= E[\pi(X)CATE(X)] - E[\pi(X)Cost(X)]
\end{aligned}
$$

- Value of targeting rule relative to targeting no one
- Recall that we know $\mathrm{CATE(X)} = E\big[\tilde{Y}\big|X\big] \Rightarrow E[\pi(X)CATE(X)] = E\big[\pi(X)\tilde{Y}\big]$
  - For any policy $\pi(X)$, we can estimate this object directly in our testing data!

Retail Spending Example

What is the estimated value of targeting (relative to treating no one)?

- Evaluate in testing data (30000 observations)
- Consider two simple scenarios:
  - Cost to target = 0
  - Cost to target = 2

	Cost = 0		Cost = 2	
Rule	Value	N. Treated	Value	N. Treated
Lasso	3.26	30000	1.95	10998
Tree	3.26	30000	0.78	1857
Forest	2.81	23135	2.08	11307
Everyone	3.26	30000	1.26	30000

Policy Learning

Given that our goal is likely to find a good policy, we can bypass estimating the $CATE$ entirely and directly try to estimate a good policy function.

We'd like to choose a policy $\pi(X)$ to maximize $V(\pi)$:

- Ignoring costs, we have $V(\pi) = E[\pi(X)CATE(X)] = E[\pi(X)\tilde{Y}]$
- $\pi(X) \in \{0,1\}$, so we can rewrite

$$\begin{aligned} E\left[\pi(X)\tilde{Y}\right] &= E\left[(2\pi(X) - 1)\tilde{Y}\right] \\ &= E\left[(2\pi(X) - 1)sign\left(\tilde{Y}\right)\left|\tilde{Y}\right|\right] \\ &= E\left[1\left((2\pi(X) - 1) = sign\left(\tilde{Y}\right)\right)\left|\tilde{Y}\right|\right] \end{aligned}$$

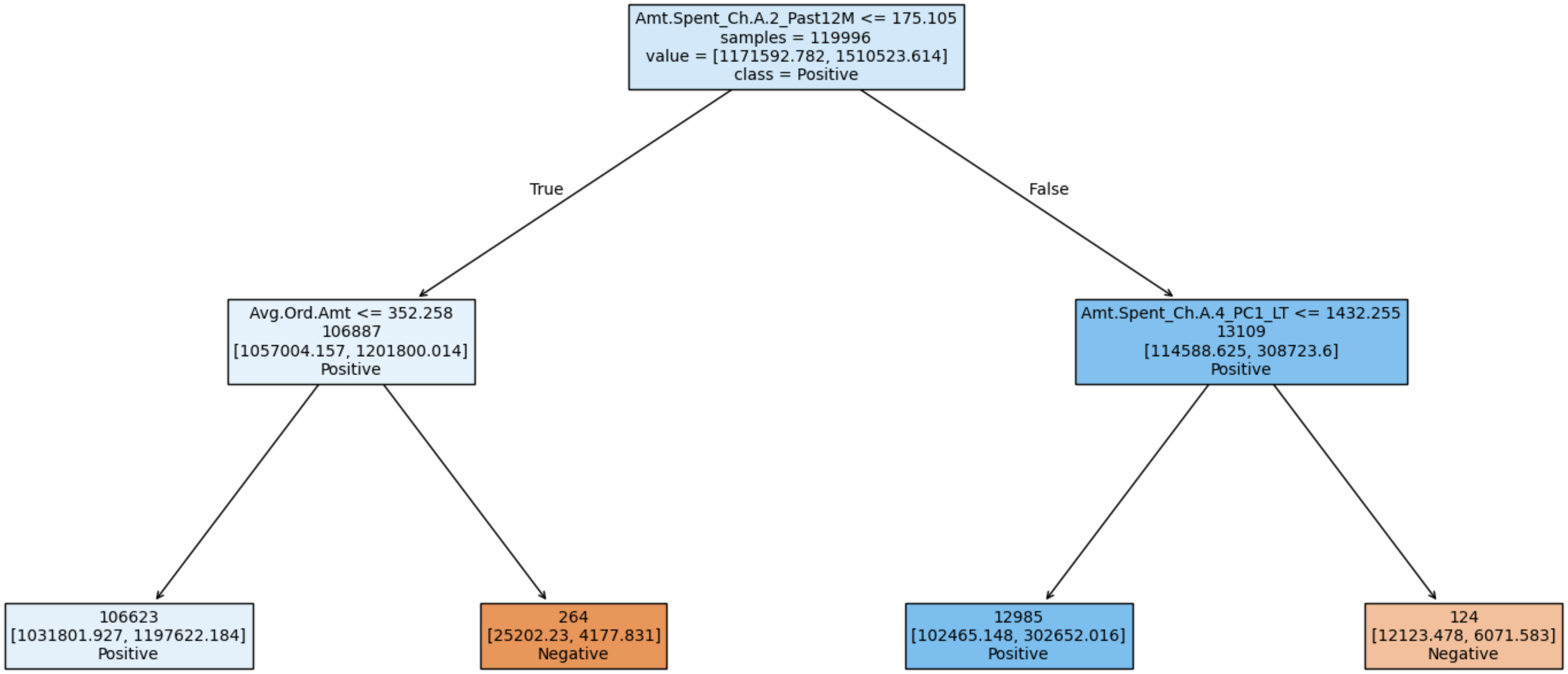
- Punchline is that policy learning is the same as classification, where our goal is correctly predict the sign of $\tilde{Y}$, but we weight by $\left|\tilde{Y}\right|$
    - Intuition is clear: Want to target where CATE(X) is positive = predict its sign
    - Get reward/loss depending on the value of CATE(X) though, so want to pay more attention to values that seem really beneficial/costly
- Means that we can use any ML classification method that allows for weights to directly estimate policy!

Retail Spending Example

What are the policy rules if we try to estimate them directly?

- Use (simple) trees because they are easy to explain and visualize

Estimated Policy (Cost = 0)



Retail Spending Example

What are the policy rules if we try to estimate them directly?

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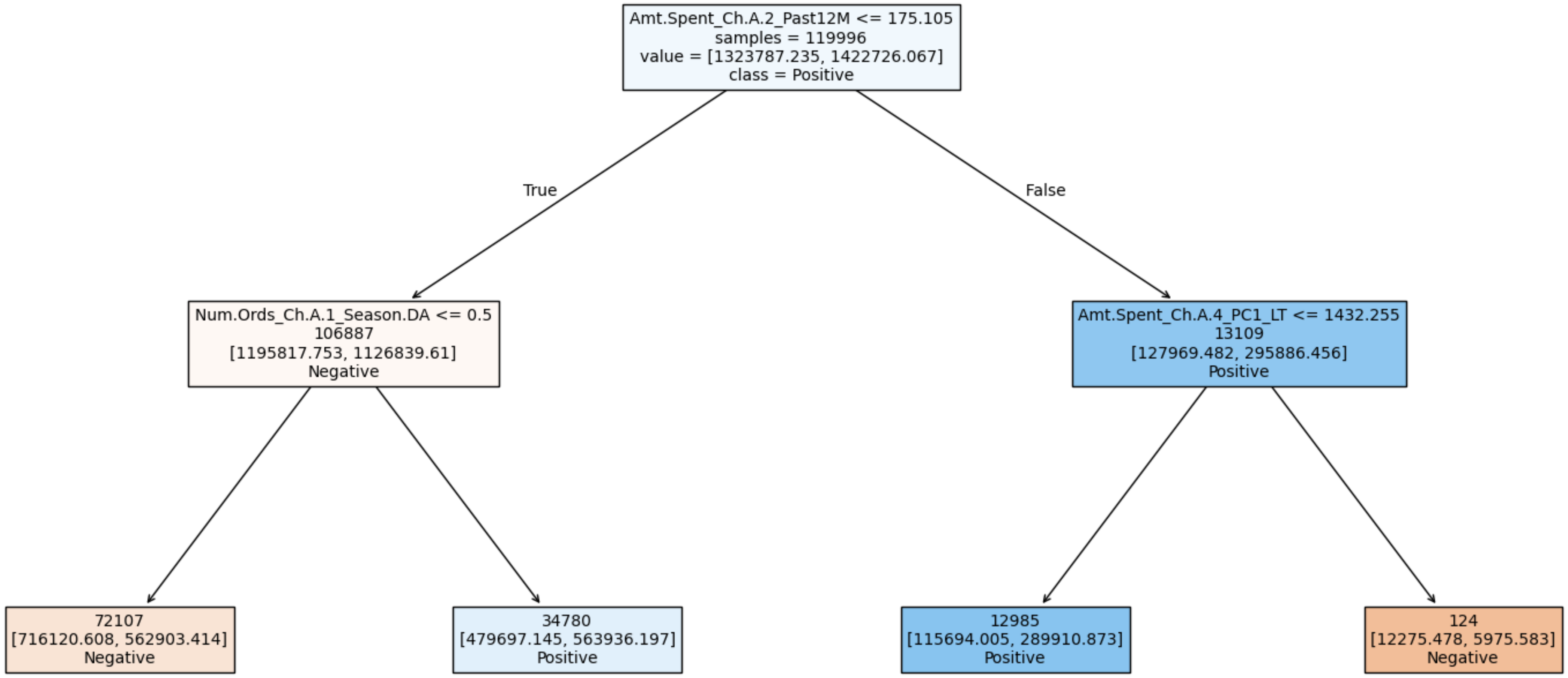
Estimated Policy (Cost = 2)



Retail Spending Example

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Everyone	3.26	30000	1.26	30000
Direct Estimate	3.21	29920	1.99	11972

4. Observational Data and (De-) Confounding

Observational Data and Causal Inference

With observational data, it is *impossible* to draw causal conclusions *without assumptions*

- some assumptions are bad/some are good: depends on the context
- seems overly nihilistic to conclude we shouldn't try to learn from observational data though
- some questions can only ever be answered with observational data
    - some experiments are impossible (or at least unethical)
- many types of assumptions available for drawing causal conclusions from observational data
- We will consider only one type of assumption

Unconfoundedness

- Selection bias is the result of variables that are related to both treatment and outcome
- These variables are observed
- After "controlling" for these variables, selection bias is removed

Unconfoundedness

Unconfoundedness

- Selection bias is the result of variables that are related to both treatment and outcome
  - Called *confounding variables* or simply *confounds*
- These variables are observed
- After "controlling" for these variables, selection bias is removed
  - Controlling is effectively making an apples-to-apples comparison
  - When we say we "control for a variable", we really mean we are making a comparison between observations *with the same value of the confound* in the treatment and control state of the treatment variable

Toy Example:

- Want to understand the causal effect of chocolate consumption on number of Nobel prize winners at the country level
  - There is even a real, hypothesized channel: flavonoids (which include chocolate) are documented to improve cognitive function
  - There is a strong correlation between country level chocolate consumption and number of Nobel prize winners
- Confounding variable: country level wealth (e.g. GDP)
  - Wealthier countries have more Nobel prize winners and also more chocolate consumption
  - When we compare countries with similar GDPs but different levels of chocolate consumption, there is no evidence of correlation between chocolate consumption and number of Nobel prize winners
- Taken from P. Maurage, A. Heeren, and M. Pesenti (2013) "Does chocolate consumption really boost Nobel Award chances? The peril of over-interpreting correlations in health studies," *Journal of Nutrition*, 143(6).

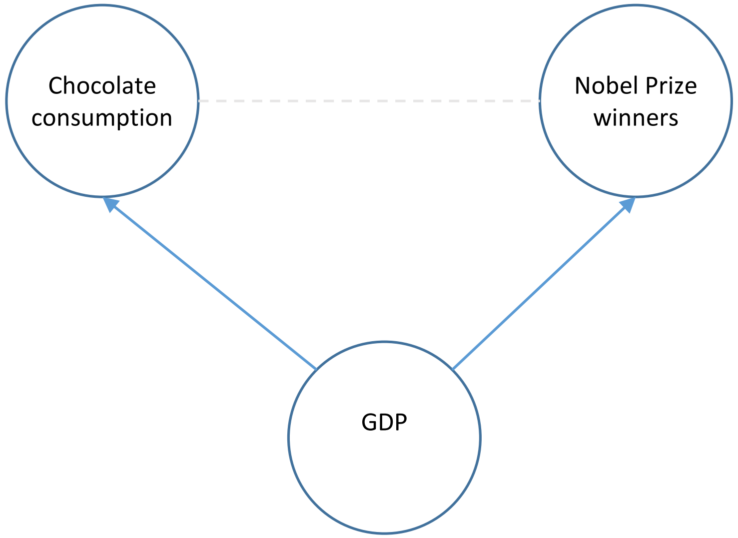
DAGs

DAG = Directed Acyclic Graph

A popular way to represent causal models that is well-suited to thinking about unconfoundedness is by drawing DAGs [Pearl (2009) *Causality*]

- A DAG is a graph
  - nodes represent variables
  - arrows connecting nodes represent connections between variables
  - direction of arrow indicates flow of "causal chain"

- Lots of rules for working with DAGs in trying to understand causal relationships
  - A simple sufficient rule for being able to learn the causal effect from one node on another is to control for all variables that cause (have arrows that point to) the nodes of interest
  - With complicated DAGs with lots of nodes and arrows in lots of directions, things get more nuanced. We're not going to worry about it in this course



DAG for chocolate/Nobel example

ML and Observational Causal Models

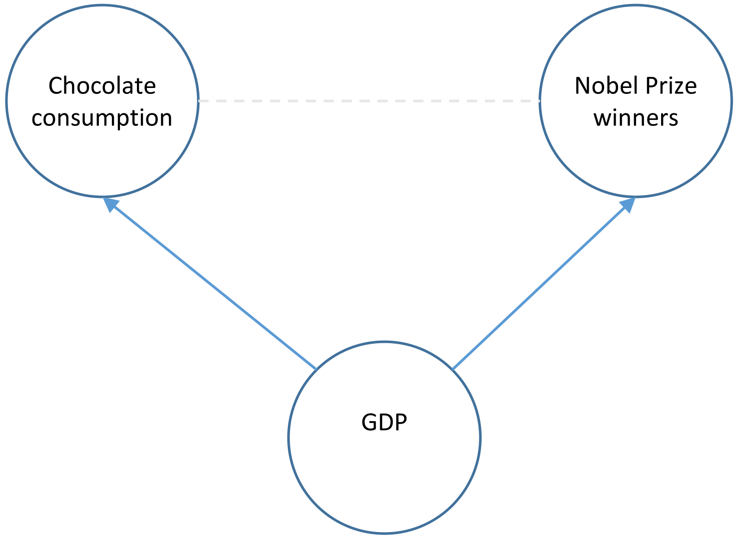
ML fits into this framework in helping us learn *how to control* for the confounding factors

Let’s look at our chocolate/Nobel example

- The DAG tells us that GDP influences both Chocolate Consumption and Number of Nobel Prize Winners
- (Heuristically) means we can only use the part of Chocolate Consumption and the part of Nobel Prize Winners that ARE NOT predicted by GDP to learn about how Chocolate Consumption and Nobel Prize winners relate to each other
- ML provides tools for
  - predicting Nobel Prize Winners with GDP
  - predicting Chocolate Consumption with GDP
  - (Heuristically) we can use the residuals from these predictions to learn about the channel we actually care about

Importantly, ML DOES NOT tell us what the confounding factors are

We need to use context and scientific knowledge to reason about the structure



DAG for chocolate/Nobel example

[Code for observational causality example]

Subscription Renewal DAG

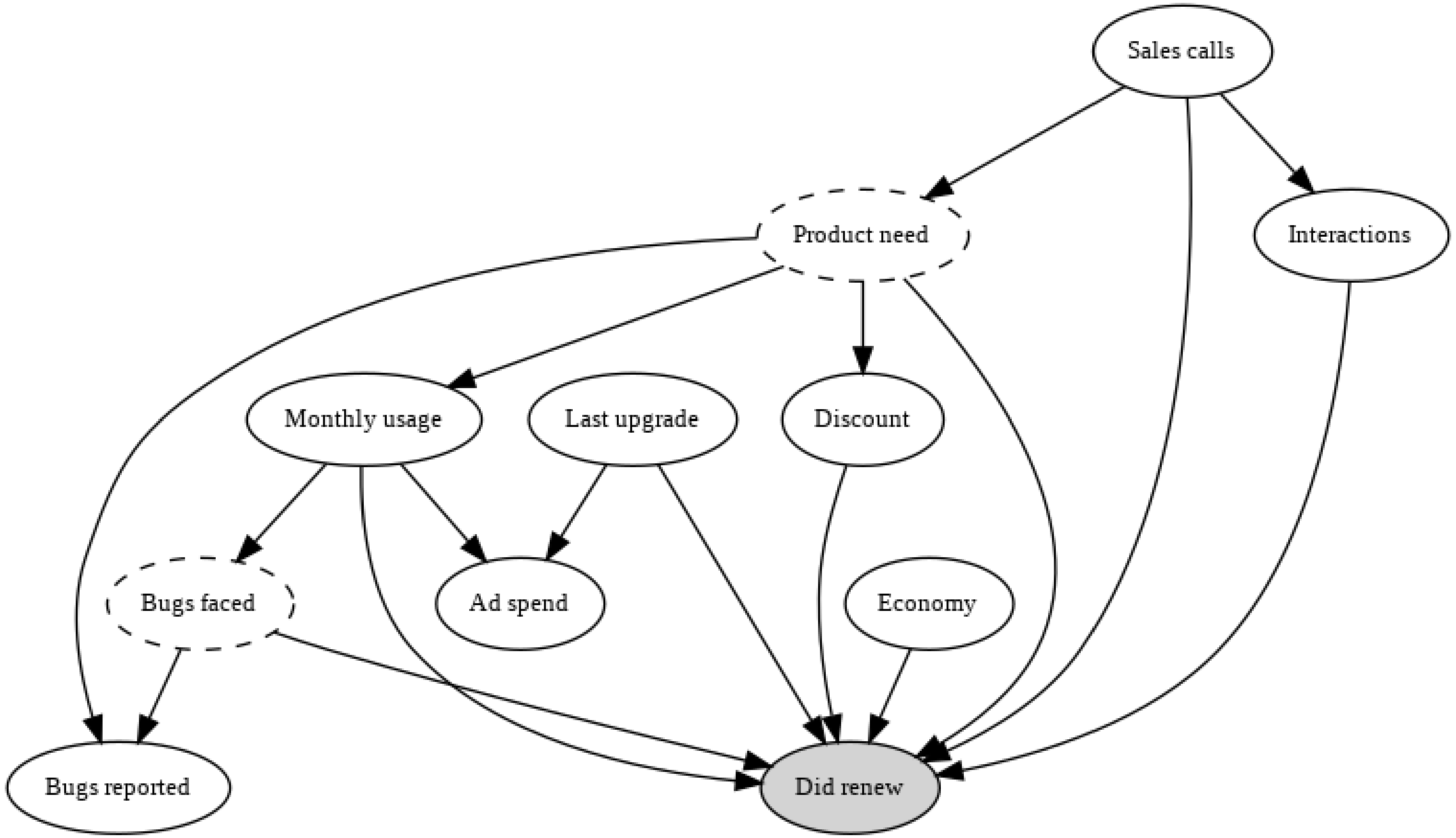
A more complicated DAG

Dashed nodes indicate unobserved/latent variables

The DAG provides a model for how the variables relate to each other (and the outcome of interest)

We *know* the DAG in this example because the data is artificial.

In real life, you need to reason about the mechanisms.



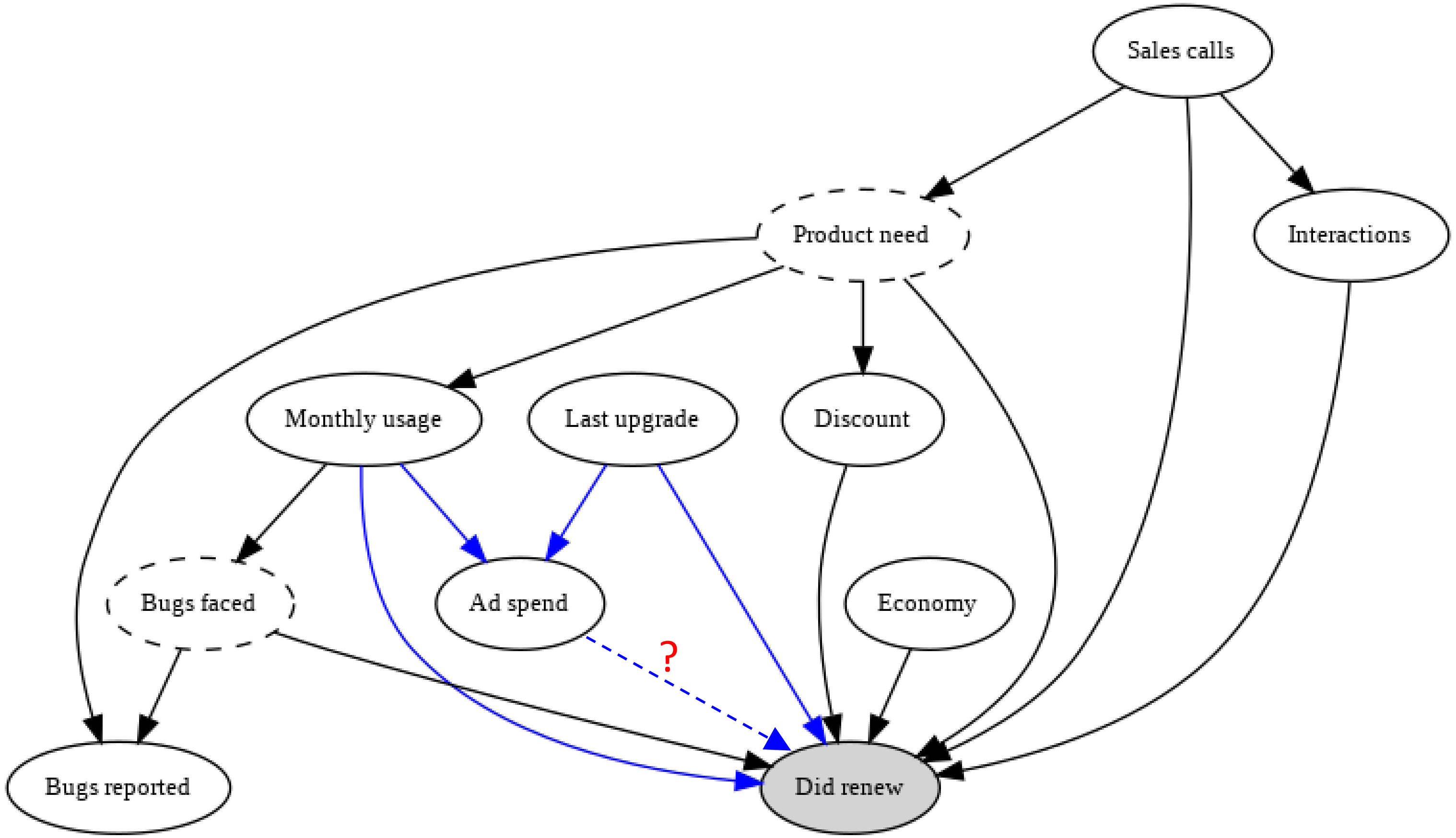
Impact of Ad Spending

Suppose we didn't know there wasn't an arrow between *Ad Spend* and *Did Renew.*

We might then want to know the causal effect of *Ad Spend* on *Did Renew.*

We see that *Ad Spend* and *Did Renew* are both impacted by *Monthly Usage* and *Last Upgrade*.

- *Monthly Usage* and *Last Upgrade* are *observed confounds*
- Maintaining the graph model, we can learn the causal effect of *Ad Spend* on *Did Renew* by *controlling for* these two variables

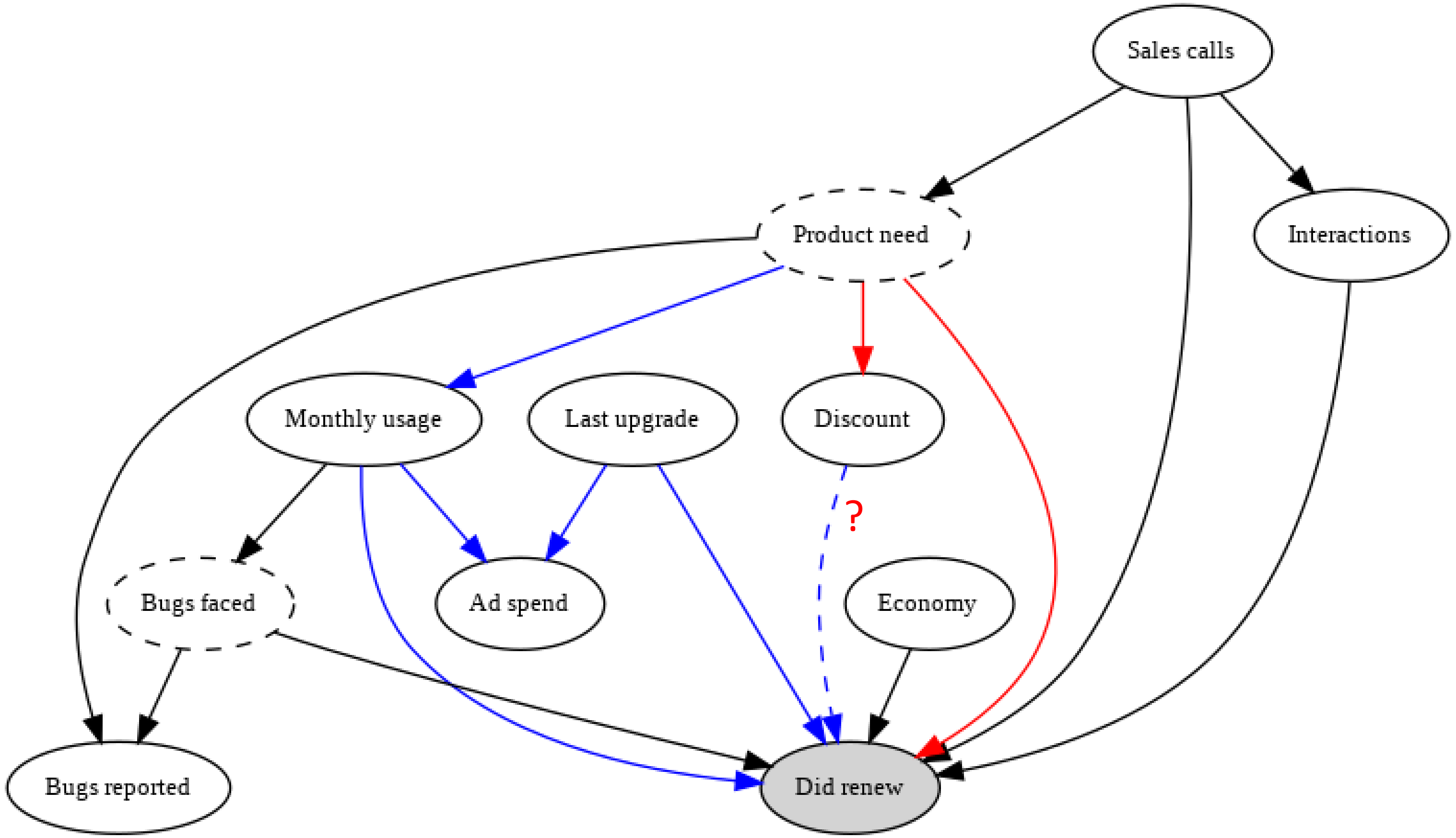


Impact of Discount

Suppose we wanted to know the causal effect of *Discount* on *Did Renew.*

We see that there are three paths connecting *Discount* to *Did Renew*

- *Monthly Usage* is an *observed confound* that blocks two of the paths
  - As an aside, we don't need to worry about the path through *Ad Spend* regardless
- *Product Need* is an *unobserved confound*
  - Because *Product Need* is unobserved, we cannot control for it
  - Means we cannot make "apples-to-apples" comparisons using the observed data and cannot learn the causal effect of *Discount* on *Did Renew*



Controlling for Confounds

Once we have decided we can learn a causal effect by controlling for observed confounding factors, how do we proceed?

- First thing we have to do is decide what we want to learn (E.g. ATE, CATE, something else)
  - Unfortunately, each potential target will require different methods

We will focus on learning a slope coefficient

- Heuristically interpreted as expected change in the outcome from *intervening* and changing the target variable by one unit
- Recall intuition that controlling for confounds means using only information that cannot be explained/predicted by the confound

Basic algorithm:

1. Build a model to predict outcome, Y, using the confounds. Obtain residuals r_Y
2. Build a model to predict treatment of interest, T, using the confounds. Obtain residual r_T
   - We still don't want to use a prediction model on the sample where it was learned. In steps 1 and 2, we do "cross-fitting" (akin to cross-validation)
3. Estimate the "causal slope" between Y and T by simple linear regression of r_Y on r_T

Impact of Ad Spending

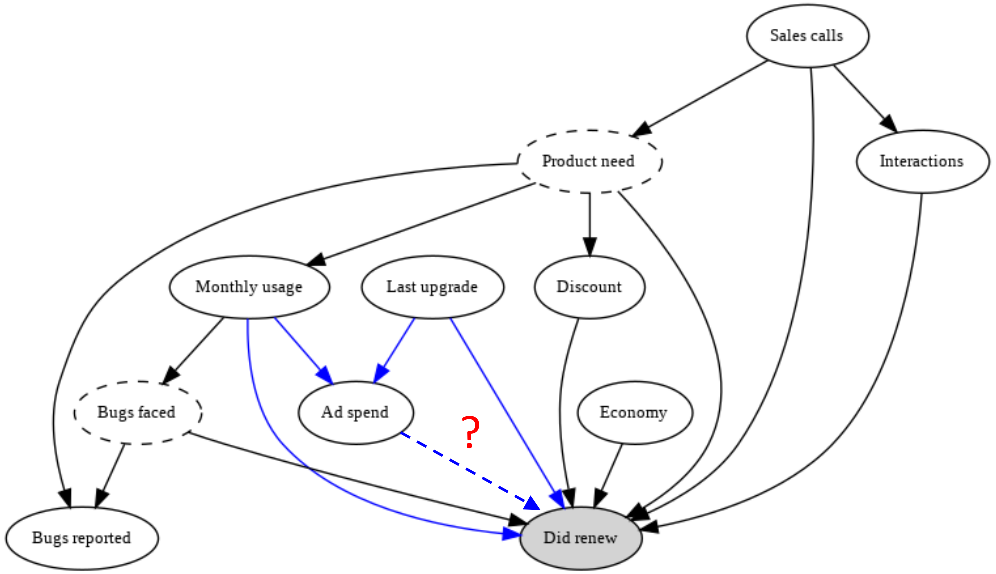
Let's estimate the causal effect of *Ad Spend* on *Did Renew*

We need to

1. build predictive model for *Ad Spend* using *Last Upgrade* and *Monthly Usage*

2. build predictive model for *Did Renew* using *Last Upgrade* and *Monthly Usage*

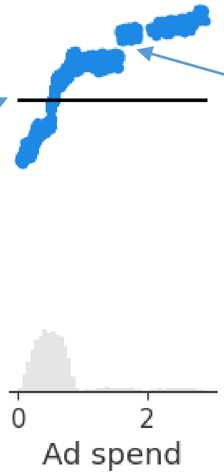
3. regress the prediction errors for *Did Renew* onto the prediction errors for *Ad Spend*



Results:

Estimated Effect: 0.107
Standard Error: 0.102

Recall that the true effect is 0.



Recall the estimated effects from the baseline predictive/observational model

5. Summary

Summary

Many business problems are fundamentally causal

- Supervised/unsupervised learning methods are fundamentally about finding patterns in data
- Without outside knowledge, patterns in data don’t say anything about causality

Knowing data from an *experiment* is ideal

- ML methods help us learn about heterogeneity and targeting

Without an experiment, can augment data with “*subject matter knowledge*” (*models*)

- DAGs are one method to encode a model
- ML methods can help us learn causal effects *within the context of the model*
    - Should think carefully about model

Could do a whole course on this topic

- E.g. causalml-book.org 😄